

The Time-Winding as a Logarithmic Spiral

The K_{frak} recursion forces self-similarity ratio e and continuous scale invariance —
and what HLV's spiral time would have to satisfy

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4 July 2026 | Dok. 295

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.1 Question: what exactly is FFGFT's time-winding?

In FFGFT, time is the mass circle, woven into the remaining circles via $\tilde{T} \cdot m = 1$ and not a separable factor (Dok. 285). A precise question follows: what exactly *is* FFGFT's time-winding?

This document settles it completely FFGFT-internally — it is *derivable* from the fractal correction. The result in one sentence: the K_{frak} recursion forces the time-winding into a *logarithmic spiral with self-similarity ratio e* , that is, into continuous scale invariance.

.2 Three building blocks from the corpus

Three quantities are fixed elsewhere and are only brought together here. First, the fractal dimension of space, $D_f = 3 - \xi_0$ with $\xi_0 = 4/30000$ (Dok. 205, Dok. 101). Second, the fractal correction in the gate: the X rotation is not by π but by $\alpha = \pi K_{\text{frak}}$ with $K_{\text{frak}} = 1 - 100\xi_0$ (Dok. 034). Third, the scale run as the recursion $\xi_{n+1} = \xi_n(1 - 100\xi_0)$, more precisely $\xi_{n+1} = \xi_n(1 - 100\xi_n)$ (Dok. 146).

From $\xi_0 = 1/7500$ follow three exact facts that carry the rest:

$$100\xi_0 = \frac{1}{75}, \quad K_{\text{frak}} = \frac{74}{75}, \quad \frac{1}{100\xi_0} = 75. \quad (1)$$

.3 Non-closure: from the 75-closure to the spiral

A winding that should advance by 2π per turn advances gate-wise only by πK_{frak} and *misses* π by $100\pi\xi_0$ per half-turn. Iterated, the winding does not close but precesses — it becomes a spiral. The closure defect per turn is $d = 100\xi_0$ in turn units.

For frozen ξ_0 , $d = 1/75$ is constant, so the rotation number $74/75$ is rational: the winding closes after *exactly 75 turns* and carries a 75-fold structure — the number $75 = 1/(100\xi_0)$ drops out exactly. This is the case of constant defect, i.e. of constant pitch. The limiting case of vanishing defect, $d = 0$ ($\xi \rightarrow 0$), by contrast closes already after one turn. Both closed cases are the *rolled-up*, compact reading of the time-winding: a single scale, the winding lies closed and is static. FFGFT already knows a non-closure algebraically, in the non-multiplicative mode composition (Dok. 193); here its geometric-temporal counterpart is added.

.4 The scale run makes the spiral logarithmic

When ξ runs over the recursion, the rational 75-closure turns into a non-closing, self-similar spiral. This transition is the *unrolling* of the winding: the spiral is the decompactified counterpart of the closed 75-winding, and “unrolling” here means precisely letting the scale run (the recursion), not merely splitting off a torus factor. The derivation is closed. With $u_n = 1/\xi_n$,

$$u_{n+1} = \frac{1}{\xi_n(1 - 100\xi_n)} = u_n(1 + 100\xi_n + \mathcal{O}(\xi_n^2)) = u_n + 100 + \mathcal{O}(\xi_n), \quad (2)$$

so $u_n \approx 100n + 7500$ and hence an *algebraic* $1/n$ decay

$$\xi_n \approx \frac{1}{100(n + 75)}. \quad (3)$$

The cumulative precession defect thereby becomes logarithmic,

$$D(k) = \sum_{n \leq k} 100 \xi_n \approx \sum_{n \leq k} \frac{1}{n + 75} \approx \ln\left(1 + \frac{k}{75}\right). \quad (4)$$

With radius $\rho = k + 75$ — the pole of the spiral sits at the effective origin $n = -75 = -1/(100 \xi_0)$, the same 75 as before — and precession angle $\beta = 2\pi D(k)$, it follows that $\rho/75 = e^{\beta/2\pi}$, that is

$$\rho = 75 e^{\beta/2\pi} = \text{logarithmic (equiangular) spiral}. \quad (5)$$

Its self-similarity is thereby fixed: per precession round, $\Delta\beta = 2\pi$, the radius scales by $e^1 = e$. The *ratio* e is not assumed but follows from the algebraic $1/n$ decay of the recursion: it is the base of the natural logarithm produced by the harmonic sum, and no additional parameter enters. The radius offset 75 is likewise not freely chosen — it is $1/(100 \xi_0)$, the same constant that already carries the 75-closure.

Both are confirmed numerically without leeway (`zeit_logspirale_probe.py`): the fit $\ln \rho = a\beta + b$ gives $a = 0.15922 \approx 1/2\pi = 0.15915$ and $e^{2\pi a} = 2.7195 \approx e$; the locally measured self-similarity ratio converges scale by scale $2.734 \rightarrow 2.726 \rightarrow 2.721 \rightarrow 2.720 \rightarrow 2.719 \rightarrow 2.719$ towards $e = 2.71828$.

.5 Continuous, not discrete scale invariance

The self-similarity is *continuous* — a log spiral, a power law — not discretely log-periodic. A test for discrete scale invariance (the residual of $D(k)$ against the smooth $\ln(1 + k/75)$) yields an amplitude near zero and no periodicity in $\ln k$. This agrees with the corpus: Dok. 146 records that the scaling over the factor $1/\xi_0 = 7500$ becomes quasi-continuous through about 13 logarithmic sub-levels, and the fractal dimension stays constant without steps. The time-winding realises the same continuous scale invariance geometrically, as a log spiral.

.6 Space and time: the same ξ_0 , scaled differently

Space and time carry the same fundamental constant ξ_0 at two places: the spatial box-dimension deficit is ξ_0 (in $D_f = 3 - \xi_0$), the winding defect per gate is $100 \xi_0 = 1/75$. The factor 100 between them is the recursion coefficient; it links a *geometric* deficit to a *dynamical* defect and privileges no axis. Which axis carries the defect is a matter of projection — it is conventionally noted on time, but is symmetric under $\tilde{T} \cdot m = 1$ (following sections), hence no amplification of time over space.

A boundary must be drawn explicitly here: the box dimension of the time-winding itself is ≈ 1 — it is a curve — and *not* $3 - \xi_0$. The link between the temporal spiral and the spatial fractal dimension is the shared ξ_0 , not an equality of dimensions. The time spiral does not carry the space dimension.

.7 The choice of axis is a projection

The logarithmic spiral is a property of the whole coupled winding on T^4 , not of a single coordinate. What is forced is the *non-closure* with ratio e ; onto which axis one places it is a matter of representation. Because $\tilde{T} \cdot m = 1$ couples time and mass inversely and the torus weaves all circles together — nothing factorises — there is no distinguished time dimension that rolls up on its own. A generic winding runs through several circles at once; the pure

time winding is the special case of a projection. The same non-closure could equally be placed on a space axis or on a mixture — different basis, same invariant.

The assignment to time is thus an artificial shift onto one axis — a convention, not a distinction. The gate defect is conventionally described on the time-winding (Dok. 034); the dual projection (next section) shows, however, that the same non-closure appears with the same factor 100 on the space side. No basis is thereby preferred; each is an equivalent representation of the same invariant — the same representational freedom as between the two equivalent representations (Dok. 206) and in the translation invariance of $\theta = 2/9$ (Dok. 293).

.8 Continuous time: the dual projection into space

If, conversely, time is set *continuous* — smooth, without gate-wise rolling up — then the non-closure can no longer sit on time; it must be carried by another axis. The coupling $\tilde{T} \cdot m = 1$ does not allow time to be smoothed in isolation: with $d\tilde{T}/\tilde{T} = -dm/m$ the mass runs along inversely, and with it the spatial-geometric side ($D_f = 3 - \xi_0$). The running winding then appears as a *variation of space* rather than as rolled-up time. It is the same invariant non-closure with ratio e , only rotated into the dual projection: in the present reading time runs (rolls up) and space stays comparable; in the dual reading time is continuous and space varies.

This mirroring has been worked out numerically (dual_projektion_probe.py). Because time advances per gate shortened by the factor $K_{\text{frak}} = 1 - 100\xi_n$, mass advances by the inverse factor $1/K_{\text{frak}}$ under $\tilde{T} \cdot m = 1$; the defect per step is $100\xi_n$ on both sides (the ratio $d_{\text{mass}}/d_{\text{time}} \rightarrow 1$ to six digits). The factor 100 therefore does not *move*, it is *mirrored*: the gate factors are inverse ($K_{\text{frak}} \leftrightarrow 1/K_{\text{frak}}$), the amplification 100 is the same. Both projections yield the same logarithmic growth, and the spiral fit gives the same ratio e for both ($e^{2\pi a} = 2.7195$ and 2.7192); their cumulative defects differ only by a *bounded* constant (≈ 0.0134 , not growing with $\ln k$). The core content is thereby representation-invariant and the non-closure preserved: time cannot be smoothed without mass and space taking over the winding — the mass-time duality principle in action. Its distribution over time, mass and space is a matter of projection; the ratio e is not.

.9 Placement relative to HLV

The FFGFT-internal statement is thereby *derived*: the K_{frak} recursion forces the time-winding to be a log spiral with ratio e , equivalent to a $1/n$ decay of its pitch. Only here does HLV enter — and here it must first be separated that “spiral time” in HLV denotes not one object but two. First the *geometric* time from the factorisation $T^7 = T^4 \times T^3$ (Dok. 285): there time is an A_5 singlet on the S^1 factor, decoupled and separable. Second the *temporal-dynamic* spiral time, the object HLV *calls* by that name: a non-Markov operator with a triadic time coordinate and a memory channel, without A_5 and without an S^1 factor. FFGFT, unlike both, does not factorise; its spiral is the projection of an inseparable whole.

The limiting case of vanishing defect ($d = 0$) noted above corresponds to the fixed point $\xi \rightarrow 0$ of the recursion, in which the defect vanishes and the winding closes. That is the *geometric* decoupled singlet time, not the temporal-dynamic object; the running FFGFT winding approaches it asymptotically. The decisive question to HLV is therefore no longer “similar?” but “ratio e , yes or no?” — and it must be put to the right one of the two objects. The following section carries out this check. Ordered within the three methodological layers: the derivation recursion $\rightarrow$ log spiral is a core derivation, obtained from ξ ; the identification with one of the two HLV times is a bridge candidate; no physical claim over HLV is made.

.10 Comparison with HLV's spiral time: shared Z_3 , no ratio e

The identification of the FFGFT log spiral with HLV's "spiral time" can now be checked against a concrete HLV source — namely against the one of the two objects designated *temporal-dynamic* in the previous section. Formally it is a non-Markov operator with a triadic time coordinate

$$\Psi(t) = (t, \phi(t), \chi(t)), \quad \Psi(t) = t + i\phi(t) + j\chi(t), \quad (6)$$

where t notes the laboratory ordering, ϕ the phase organisation and χ a memory; it carries neither A_5 nor a singlet nor an S^1 factor and possesses a memory kernel K_χ with Markov limit $g_\chi \rightarrow 0$. Precisely this object HLV calls *spiral time*; the decoupled A_5 singlet (previous sections) is the other. The two share only the name.

.11 The temporal-dynamic object does not carry the ratio e

The test question posed above — "ratio e , yes or no?" — targets a geometrically self-similar object. The temporal-dynamic object is not of that category. It is rather an object of open quantum dynamics: its total dynamics is — per Marcel's spiral-time paper — a unitary dilation on the product space $H_S \otimes H_M$ of system and memory, and it is tested there by open-system criteria (process tensor, CP-divisibility, hard reset, finite-bath comparison), not by a geometric self-similarity. It thereby belongs recognisably to the dynamical, not the geometric category; the e -question is put to an object that, by its own construction, makes no such claim. It carries no derived self-similarity ratio, no $1/n$ decay of a pitch and no scale run; its non-closure sits in the memory channel χ , not in a running scale. Its only self-similar internal structure is the golden-ratio / phason order of the associated cut-and-project (the control replaces the φ structure by random irrational phases), that is φ and not e . On the e -criterion the identification is therefore not supported. What both frameworks structurally share is the role of non-closure as the temporal core object and an explicit memory-off limit: FFGFT's case of constant defect ($d = 1/75$ fixed for frozen ξ_0 ; bounded window, linear $D(k)$) corresponds to Marcel's Markov fallback $g_\chi \rightarrow 0$. Same role, different mechanism — geometric accumulation ($D(k) = \sum 100 \xi_n \rightarrow \ln$) against a stochastic memory kernel.

FFGFT's non-closure is available not only geometrically as a log spiral but equivalently in the Hilbert space: as a memory kernel, the Fourier transform of the discrete spectral density $\sum_k g_k^2 \delta(\omega - \omega_k)$ of the T^4 connection-Laplacian (Dok. 283) — the same inseparable object in two representations ($H = L^2(T^4) \otimes \mathbb{C}^3$, a lossless bijection, Dok. 230/282); this is representational robustness, not additional evidence. The comparison can therefore be run in Marcel's own language, kernel against kernel: FFGFT's kernel is oscillatory and discrete-spectral (revivals, BLP backflow 5.125), Marcel's decaying and bounded — the same object class, a different form (Dok. 283). Precisely the $1/n$ scale run that carries e is what the decaying kernel lacks.

.12 Computational check of the e -criterion

The e -criterion is necessary and sufficient via the pitch decay: a winding is equiangular (a log spiral, ratio e per 2π) exactly when the defect per step decays like $1/n$ and the cumulative defect grows logarithmically. A constant defect, by contrast, gives an Archimedean winding with ratio $\rightarrow 1$. Both constructions can be measured against this on their own definitions (spiralzeit_verhaeltnis_probe.py).

FFGFT carries e : with $d_n = 100 \xi_n$ one has $d_n(n + 75) \rightarrow 1$ — the $1/n$ decay — and the spiral fit gives $e^{2\pi a} = 2.71947 \approx e$. Marcel's spiral time does not: its phase $\phi_i = q_\phi u_i$ is linear

in the internal coordinate and its memory $\chi_i = F(\rho_i)$ is a bounded window; the defect per step stays constant, $D(k)$ grows linearly rather than logarithmically ($D(10^5) = 1333$ against $\ln(1 + k/75) = 7.2$), and the measured ratio is 1.0011. On its own definitions it is thus not a log spiral at all, but a winding of constant pitch.

The geometry HLV actually carries points, moreover, to a constant other than e . Were one to build an equiangular spiral from the $\tau = \varphi$ cut-and-project, its natural self-similarity factor would be golden — $\varphi = 1.618$ per turn, or $\varphi^4 = 6.854$ per 2π — not $e = 2.718$. Both frameworks thus know, in so far as they carry self-similarity at all, different constants: e on the FFGFT side (from the $1/n$ decay of the ξ recursion), φ on the geometric HLV side (from the cut-and-project). As a bridge condition in the sense of P35 this reads precisely: HLV's spiral time carries e only if its phase law contains a $1/n$ scale run — a running internal scale —; the given definitions do not contain it. That is a candidate, not a derivation.

.13 Basis and dimensions: same numbers, different meaning

Whether both frameworks carry the *same* basis can be ordered dimension by dimension. The numbers coincide, the meaning of the dimensions does not. The *three*: FFGFT's three is the internal $\mathbb{C}^3$, i.e. the Z_3 order on T^4/Z_3 (Dok. 282) — a mode and generation index over the torus. Marcel's three is the triadic *time* (t, ϕ, χ) — a decomposition of the one ordering axis into order, phase and memory. In his root-mass note this three coincides with a C_3/Z_3 orbit ($2\pi k/3$, $k = 0, 1, 2$); to that extent both frameworks share the Z_3 ($C_3 < A_5$, Dok. 285). What is shared is the Z_3 , not a common three-dimensional coordinate space: one indexes modes, the other time roles.

The *six*: FFGFT's six is not native. It appears only via the HLV embedding $T^7 = T^4 \times T^3$ (Dok. 285); FFGFT itself is four-dimensional (T^4). Marcel's six is the parent lattice $\mathbb{Z}^6$ of the cut-and-project, split into three physical directions ($P_{\parallel}$) and three internal, phason ones ($P_{\perp}$). The same number, a different anatomy: a $4 + 3$ torus embedding against a $3 + 3$ cut-and-project. Moreover, in Marcel's case the six is not the basis of the *continuous* spiral time — that one is triadic (the three) — but of the *discrete* G lattice; the two levels are for him explicitly representations of the same operator, not a replacement of it. This corresponds to the representational freedom on the FFGFT side (two equivalent representations, Dok. 206; the rolled-up 75-winding against the unrolled log spiral).

The only concretely established overlap is the φ -icosahedral shared object ($C_3 < A_5$), computationally established but *one-sided* — the test direction is FFGFT→HLV (Dok. 294) — and filtration-sensitive: it separates HLV sharply from crystalline controls but only weakly from a generic $\sqrt{2}$ cut-and-project. Computationally this can be pinned to the axis spectrum (basis_6d_achsen_probe.py): the six projected axes of the 6D cut-and-project are equiangular only at $\tau = \varphi$ — all fifteen pairwise angles equal to 63.435° — while $\tau = \sqrt{2}$ gives a two-valued spectrum ($61.87^\circ/70.53^\circ$). The six therefore do not coincide per se; what is equal is the golden-icosahedral projection alone. Basis equality is thus a *candidate*, not a derived identity. The numbers 3 and 6 coincide, the shared invariant is the Z_3 , and the dimensions themselves carry different meanings.

.14 Status

Secured and derived: the time-winding as a logarithmic spiral with self-similarity ratio e , continuous scale invariance, from the K_{frak} recursion; the exact 75-closure in the frozen case; the shared constant ξ_0 , appearing as the spatial box deficit ξ_0 and as the winding defect $100\xi_0$ (factor 100 = recursion coefficient, axis-symmetric). The closed 75-winding is here

the rolled-up (compact, single-scale) reading, the log spiral the unrolled (decompactified, across-scales) reading of the same time-winding.

The comparison with HLV's spiral time is negative on the e -criterion: the temporal-dynamic object carries no ratio e , but golden-ratio / phason quasiperiodicity; the basis question reduces to the shared Z_3 ($C_3 < A_5$, one-sidedly verified, filtration-sensitive). The sharp residual question shifts to the *geometric* HLV object — whether the factorised layer (A_5 singlet on S^1) knows a scale-invariant object that would be testable against e —; in the available sources no such object is exhibited. The category separation (two spiral times) and the dimension-by-dimension comparison are secured findings; the identification remains a bridge candidate, currently blocked on the e -criterion.

The contrast here is structural, not ontological. Both frameworks — and likewise the two variants within FFGFT, whether time or space is rolled up — are mathematical models, not reality; no factorisation and no choice of axis is thereby ontologically privileged. That one hopes such a model is applicable to reality is the motivation, not a statement about what is.

Reproducible in `zeit_windung_probe.py`, `zeit_logspirale_probe.py`,
`dual_projektion_probe.py`, `spiralzeit_verhaeltnis_probe.py` and
`basis_6d_achsen_probe.py`.